

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Supported the delivery of continuous improvement initiatives by applying **process mapping** and **root cause analysis** to identify waste reduction opportunities within a regulated food manufacturing environment.
- Collected and analysed **operational data** using **SQL** and scripting to track performance metrics and support evidence-based decision making for sustainability objectives.
- Collaborated with **cross-functional teams**, including operations and quality assurance, to embed **Lean ways of working** into daily production activities.
- Assisted in identifying improvement opportunities that enhance **sustainability**, energy efficiency, and responsible resource use within manufacturing processes.
- Developed **Python scripts** to automate data collection and reporting for key performance indicators related to waste reduction and environmental impact, reducing manual effort by an estimated **15%**.
- Participated in **sprint planning** and Agile ceremonies, contributing to the estimation of tasks and refinement of user stories for process improvement projects.
- Containerised data processing applications using **Docker** and deployed them to **AWS EC2** instances, ensuring consistent environments for analytical workloads.

SKILLS

Lean & Continuous Improvement – Process Mapping, Root Cause Analysis, Standard Work, Waste Reduction, Continuous Improvement Workshops, Visual Management

Data Analysis & Reporting – SQL, Python, Pandas, NumPy, R, Data Collection, Performance Tracking, Business Intelligence

Sustainability – Environmental Performance, Energy Efficiency, Waste Reduction, Resource Management

Software Development & Tools – Java, Spring Boot, REST APIs, Python Scripting, Git, Jenkins, Jira, Confluence, Docker, AWS

Food Manufacturing & Compliance – Food Safety, Quality Assurance, Good Manufacturing Practices (GMP), Health & Safety Standards

PROJECTS

Operational Efficiency Analysis for Food Production Lines

- Analysed production line data using **Python** and **Pandas** to identify bottlenecks and inefficiencies, focusing on reducing cycle times.
- Developed **SQL queries** to extract historical production data from relational databases, enabling detailed performance trend analysis over a **6-month period**.
- Created a series of **operational dashboards** using Power BI to visualize key performance indicators, facilitating data-driven discussions with operations management.

Energy Consumption Monitoring and Optimization

- Implemented automated data collection scripts in **Python** to gather energy usage metrics from various production machinery.
- Mapped energy consumption patterns against production schedules, identifying peak usage periods and potential areas for **energy efficiency** improvements.
- Collaborated with the facilities team to propose adjustments to machine operating schedules, aiming to reduce peak load by approximately **8%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)